

Chapter-1 Permutation and Combination

• Basic Principle of Counting

① Multiplication Rule

- If an event E_1 can occur n_1 number of ways, following which an event E_2 can occur n_2 number of ways, then total number of ways that events E_1 and E_2 occur is $n_1 \times n_2$.

e.g: @ How many 2-different digits number can be formed by using 3, 4, 5, 6?

Soln.

T O
 $\boxed{3 \times 4} \rightarrow \text{No. of choices}$

$= 12$

Unit place can be filled up by 4 ways

Tens place can be filled up by 3 ways

By basic principle of counting,

Total no. of ways = $4 \times 3 = 12$

② Addition Rule

- If an event E_1 can occur n_1 number of ways, following which an event E_2 can occur n_2 number of ways, then total number of ways that either events E_1 or E_2 occur is $n_1 + n_2$.

e.g: @ There are 4 bus routes and 5 train routes from A to B. In how many ways can a person travel from A to B.

Soln A person can travel from A to B by using either Bus or Train
 He/she can use bus by 4 different ways
 He/she can use train by 5 different ways.

$$\therefore \text{Total Number of ways} = 4 + 5 = 9$$

Exercise - 1.1

③ How many numbers of three different digits can be made by using 1, 2, 3, 4, 5, 6? How many of them are even?

→ Given, 1, 2, 3, 4, 5, 6

1st case: Unit place can be filled up by 6 ways

Tens place can be filled

up by 5 ways

Hundreds place can be filled up by 4 ways

H T O

$$4 \times 5 \times 6$$

$$= 120$$

For even

H T O

$$4 \times 5 \times 3$$

$$= 60$$

$\therefore$ Required no. of 3-different digits = $6 \times 5 \times 4 = 120$

2nd case

To form even numbers, unit place must have even digit i.e. 2, 4, 6

unit place can be filled up by 3 ways

Tens place can be filled up by 5 ways

Hundreds place can be filled up by 4 ways

$\therefore$ Required no. of 3-diff. digits = $3 \times 5 \times 4 = 60$

- ⑤ How many positive numbers less than 100 are possible by using the digits 0, 1, 2, 3? If the repetition of the digits is not allowed, how many such numbers are possible?

Soln.

1st Part

Given: 0, 1, 2, 3

We have to form positive numbers less than 100, which can be either of 1-digit or 2-digits.

no. of 1-digit +ve numbers = 3

no. of 2-digit +ve numbers = $3 \times 4 = 12$

1 digit = 3

2 digit $\rightarrow$ T 0 $\rightarrow + = 15$

$$3 \times 4 = 12$$

Repetition is not allowed

1-digit = 3

2-digit = T 0 $\rightarrow + = 12$

$$3 \times 3 = 9$$

$\therefore$ Required no. of positive numbers less than 100 = $3 + 12 = 15$

2nd Part

If the repetition of the digits is not allowed,

no. of 1-digit +ve numbers = 3

no. of 2-digit +ve number = $3 \times 3 = 9$

$\therefore$ Required no. of positive numbers less than 100 Without repetition of the digits = $3 + 9 = 12$

- ① There are 5 air flights and 15 buses per day to travel from Bhairahawa to Kathmandu. In how many ways can a man travel from Bhairahawa to Kathmandu per day?

Ans. A man can travel from Bhairahawa to Kathmandu by using either air flights or buses. He can use flights by 5 ways and He can use buses by 15 ways per day

$\therefore$ Total no. of ways $= 5 + 15 = 20$ ways.

- ② There are 5 routes from station A to B and 4 routes from station B to C. Answer the questions

- a) How many different routes are possible from station A to station C?

Ans. Routes from station A to B $= 5$
Routes from station B to C $= 4$

$A \equiv B \equiv C$

$$5 \times 4$$

$$= 20$$

Routes from A to C $=$ Routes from station A to B and Routes from station B to C $= 5 \times 4 = 20$.

- b) In how many ways can a person travel from A to C and come back from C to A?

Ans. Routes from A to C $= 5 \times 4 = 20$

Routes from C to A $= 4 \times 5 = 20$

$C \equiv B \equiv A$

$$4 \times 5 = 20$$

$$\Rightarrow 20 \times 20 = 400$$

A person can travel from A to C and come back from C to A in 20×20 ways
 $= 400$ ways.

c) In how many ways can a person travel from A to C and return back to A by using different routes?

Ans.

A person can travel from A to C in 20 ways

A person can travel from C to A using different routes = $3 \times 4 = 12$



$$A \text{ to } C = 5 \times 4 = 20$$

C to A using different routes

$$= 3 \times 4 = 12$$

Now, A person can travel from A to C and return back to A by using different routes in 20×12 ways
 $= 240$ ways.

④ How many 4 digit numbers are possible by using the digits 0, 1, 2, 3, 4, 5? If (a) the repetition of the digits is not allowed? (b) the repetition of the digits is allowed?

Soln, Given, 0, 1, 2, 3, 4, 5

a) We have to form 4 digit numbers so, thousands place shouldn't contain 0.

$$\begin{array}{cccc} \text{(a)} & T & H & T & O \\ & 5 & \times & 5 & \times & 4 & \times & 3 \\ & = & 300 & \text{ways} \end{array}$$

Thousands place can be filled by 5 ways

Hundreds place can be filled by 5 ways

Tens place can be filled by 4 ways

Ones place can be filled by 3 ways

$$\begin{array}{cccc} \text{(b)} & T & H & T & O \\ & 5 & \times & 6 & \times & 6 & \times & 6 \\ & = & 1080 & \text{ways} \end{array}$$

$$\therefore \text{Required no. of 4 digit numbers} = 5 \times 5 \times 4 \times 3 = 300 \text{ ways}$$

b) If Repetition is allowed,

Thousands place can be filled by 5 Ways

Hundreds place can be filled by 6 Ways

Tens place can be filled by 6 Ways

Ones place can be filled by 6 Ways

$\therefore$ Required no. of 4 digits number $= 5 \times 6 \times 6 \times 6$

$= 1080$ Ways

⑥ How many even numbers are possible between 2000 and 3000 by using the digits 0, 1, 2, 3, 4, ~~5~~ only once?

Soln,

Given: 0, 1, 2, 3, 4

To form even numbers

between 2000-3000,

Th H T O

$1 \times 2 \times 3 \times 2$

$= 12$

Thousands place must have

digit '2'. and Unit place must have even digit. So,

Thousands place can be filled up by 1 way

Ones place can be filled up by 2 ways

Tens place can be filled up by 3 ways

hundreds place can be filled up by 2 ways.

$\therefore$ Required no. of even numbers between

2000-3000 $= 1 \times 2 \times 3 \times 2$

$= 12$

ways

⑦ How many numbers of three digits can be formed from the integers 2, 3, 4, 5, 6 if repetition of digits is allowed? How

→

many of them will be divisible by 5?

Soln,

Given, 2, 3, 4, 5, 6

For three digits numbers

Unit place can be filled by
5 Ways

Tens place can be filled by 5
Ways

Hundreds place can be filled by
5 Ways.

H T O

$5 \times 5 \times 5$

$= 125$

divisible by 5

H T O

$5 \times 5 \times 1$

$= 25$

$\therefore$ Required no = $5 \times 5 \times 5 = 125$

Again,

For three digits number divisible by 5, ~~once~~

unit place must have digit '0' or '5'. so,

unit place can be filled by 1 way

Tens place can be filled by 5 ways

Hundreds place can be filled by 5 ways

Now,

Number of three digits number divisible by

5 are $1 \times 5 \times 5$

$= 25$ ways $\neq$

⑧ How many even numbers of at most 2-digits can be formed by using the digits 1, 2, 3, 4, 5 only once?

Soln,

Given: 1, 2, 3, 4, 5

We can form either

1-digit or 2-digits even numbers.

1-digit $\Rightarrow 2$

OR

2 digits $\Rightarrow$ T O

4×2

$= 8$

Here, 1-digit even numbers are 2 & 4.

$\therefore$ no. of 1-digit even no $= 2$

For 2-digit even numbers, unit place must have even digit. So, unit place can be filled up by 2 ways.

Tens place can be filled up by 4 ways

2-digits even number $= 2 \times 4 = 8$

$\therefore$ Required numbers $= 2 + 8$

$= 10$